

DSA:: Mock End Sem (2024-25) Sem2 : Question Bank

Q. No.	BL	CO	Questions	Unit No	Marks Weightage
			UNIT III		
1	2	3	Demonstrate Deletion Operation in AVL with example	3	6
2	3	3	Draw any directed graph with minimum 6 nodes and represent graph using adjacency matrix, adjacency list and adjacency multi list.	3	6
3	2	3	Find out the minimum spanning tree of the following graph figure 2 using : (i) Prim's Algorithm (ii) Kruskal's Algorithm.	3	6
4	3	3	Write a 'C++' function to find the shortest path in a graph using Dijkstra's algorithm with an example.	3	6
5	1	3	Define the following w.r.t. graphs with examples :(i) Degree of node (ii) Isolated node (iii) Path (iv) Cycle.	3	6
6	3	3	Write pseudo-code for Kruskal's method for finding MST of graph and explain the same for the graph whose weight matrix is given in figure	3	6
			UNIT IV		
7	3	5	Build AVL tree for given sequence of data. Show balance factor of all nodes and name the rotation used for balancing the tree 40,60,80,50,45,47,44,42,75,46,41.	4	6
8	2	5	What is OBST? List binary search tree with 3 words (w1, w2, w3) = (do, if, stop) words occurs with probabilities (P1, P2, p3) = (0.4,0.5,0.1) find expected access time in each case.	4	6
9	2	5	Write a short note on Splay Trees.	4	5
10	2	5	Explain following terms w.r.t. height balance tree LL, RR, LR, RL	4	5
11	2	5	Demonstrate Deletion Operation in AVL with example	4	6
12	2	5	Explain with example i) Red - Black Tree ii) Splay Tree	4	6
			UNIT V		
13	3	5	Build B+ tree of order 3 for the following data: F, S, Q, K, C, L, H, T, V, W, M, R	5	6
14	2	5	What is B+ tree? Give structure of it's internal node. What is the difference between B and B+ tree.	5	6
15	3	5	Construct the B+ Tree of order 4 for the following data: 1, 4, 7, 10, 17, 21, 31, 25, 19, 20, 28, 42.	5	6
16	2	5	Write an algorithm of B tree deletion.	5	6
17	2	5	Write an algorithm to delete a node from B+tree.	5	6
18	2	5	Explain with example trie tree. Give advantage and applications of trie tree	5	6
			UNIT VI		
19	2	6	What is file? List different file opening modes in C++. Explain concept of inverted files.	6	6
20	2	6	Explain any 3 operations carried out on sequentail file and its pseudo code.	6	6

21	1	6	Define sequential file organization. Give it's advantages and disadvantages.	6	5
22	2	6	Write short notes on: i) Factors affecting the file organization ii) Indexed sequential files iii) Indexing technique	6	6
23	2	6	A write a C++ program to create a file. Insert records into the file by opening file in append mode. Search for a specific record into file.	6	6
24	1	6	Define external sort with eaxmple.	6	5